Middle East Technical University
Northern Cyprus Campus
Computer Engineering Program

CNG 476 - System Simulation
Final Project Report Submission Guidelines

Spring 2025-2026

Course Code	CNG 476
Course Title	System Simulation
Instructor	Asst. Prof. Dr. Muhammad Toaha R. Khan
Email	khan@metu.edu.tr
Semester	Spring 2025-2026
Submission	Final report PDF, public GitHub repository, and final demo

1 Purpose of the Final Submission

Each group must submit a complete final project report for the selected CNG 476 System Simulation project. The report must clearly explain the implemented simulation study, including the system model, assumptions, OMNeT++ implementation, INET integration, FLoRa integration, experimental design, results, and statistical analysis.

The final project must be in **working condition**. The submitted simulation project must compile, execute, and produce meaningful simulation outputs that are consistent with the results reported in the final report. A project that is only described in the report but does not run correctly will be considered incomplete.

The final project must demonstrate meaningful use of the course topics, including discrete-event simulation, random number generation, probability models, stochastic processes, queueing analysis, input analysis, output analysis, and IoT network simulation.

The final report must contain the required tables specified in this document. Each table must have a proper caption, must be referenced in the text, and must contain project-specific information.

2 Overall Project Grade Distribution

The project grade is distributed as follows:

This document defines the expectations for the final report, GitHub repository, and final demo.

Table 1: Project Grade Distribution

Component	Weight
Interim Project Proposal Report	20%
Final Project Report	50%
GitHub Repository Submission	10%
Final Demo and Oral Evaluation	20%
Total	**100%**

3 Submission Requirements

Each group must submit:

1. The final project report as a PDF file on ODTU CLASS.
2. A public GitHub repository containing the complete project source code.
3. The final report PDF inside the GitHub repository.
4. The public GitHub repository link inside the final report.
5. A final demo during the assigned demo session.

The submitted project must be complete and executable. The GitHub repository must contain all files required to compile, configure, run, and reproduce the main simulation results. The report, repository, and demo must refer to the same implemented project. Mismatch between the report and the submitted code may result in grade reduction.

The final report submitted to ODTU CLASS will be checked through Turnitin for text similarity and AI-writing indicators. The maximum accepted threshold is **30%**. Reports exceeding this threshold may be subject to additional oral inspection, grade reduction, or academic-integrity procedures.

4 Required Final Report Format

The final report must be prepared in LaTeX using the IEEEtran conference format:

```
\documentclass[conference]{IEEEtran}
```

The report must be written in English and must contain properly numbered sections, figures, tables, equations, and references. The recommended length is **8-10 pages**, excluding references and appendices.

Each required table must be completed using project-specific information. Empty, generic, copied, or irrelevant tables will not be accepted as valid evidence.

The report must include a proper references section. The minimum number of valid references is **15**. These references must be relevant to the selected project, simulation methodology, OMNeT++, INET, FLoRa, LoRa/LoRaWAN communication, IoT application domain, probability modelling, queueing analysis, input analysis, output analysis, or performance evaluation.

The references must be cited inside the text. References listed at the end but not cited in the report will not be counted as valid references.

5 Required First-Page Information

The first page of the final report must include:

1. Project title.
2. Course code and title: **CNG 476 - System Simulation**.
3. Semester: **Spring 2025-2026**.
4. Instructor: **Asst. Prof. Dr. Muhammad Toaha R. Khan**.
5. Group number, if assigned.
6. Full names and student IDs of all group members.
7. Public GitHub repository link.
8. Date of submission.

6 Required Chapters of the Final Report

The final report must contain the following chapters/sections.

6.1 Abstract

The abstract must briefly summarise the project problem, simulated system, tools used, main performance metrics, and key findings.

6.2 Introduction

The introduction must explain the selected IoT application scenario, the motivation of the problem, the need for simulation, and the objectives of the project.

6.3 Selected Project Description

This section must clearly describe the selected project topic, the real-world system being modelled, and the purpose of the simulation study.

This section must include **Table 1: Selected Project Summary**. The table must contain the selected project title, application domain, main problem, simulation objective, main users or entities, and main performance metrics.

Table 2: Selected Project Summary

Item	**Description**
Selected Project Title	
Application Domain	
Main Problem	
Simulation Objective	
Main Users/Entities	
Main Performance Metrics	

6.4 System Architecture

This section must include at least one technical figure showing the system architecture. The figure must show the main entities such as sensor nodes, LoRa nodes, gateways, network server, application server, controller, or end users, depending on the selected topic. The figure must be explained in the text.

6.5 System Model and Assumptions

This section must define the simulation model. It must describe the system entities, events, state variables, communication flow, queueing points, traffic generation model, wireless communication assumptions, simulation assumptions, and simplifications.

This section must include **Table 3: Simulation Entities, Events, and State Variables**. The table must identify the main simulation entities, the events associated with each entity, and the state variables maintained or updated during the simulation.

Table 3: Simulation Entities, Events, and State Variables

Entity	**Events**	**State Variables**

This section must also include **Table 4: Modelling Assumptions and Justification**. The table must state each major modelling assumption and provide a brief technical justification.

Table 4: Modelling Assumptions and Justification

Assumption	**Justification**

Table 5: Random Processes and Probability Models

Process	Distribution/Model	Purpose in the Simulation
Packet generation		
Sensing event		
Failure/alarm event		
Service time		

6.6 Simulation Methodology

This section must explain how the project is modelled as a discrete-event simulation. It must describe the event scheduling logic, random number generation, probability distributions, stochastic processes, Monte Carlo repeated runs, and any time-stepped or trace-driven aspects.

This section must include **Table 5: Random Processes and Probability Models**. The table must specify each random process, the distribution or model used, and the purpose of that process in the simulation.

This section must also include **Table 6: Scheduled Events in the Discrete-Event Simulation**. The table must list each scheduled event, the trigger condition, and the resulting state change or action.

Table 6: Scheduled Events in the Discrete-Event Simulation

Scheduled Event	Trigger Condition	State Change / Action

6.7 OMNeT++, INET, and FLoRa Implementation

This section must explain the actual implementation. It must describe the OMNeT++ project structure, NED files, C++ files, message files, INI configuration files, INET components, FLoRa components, packet/message flow, and evidence that the project compiles and runs correctly.

Students must clearly explain the implemented files and modules rather than only mentioning their names. The submitted implementation must be functional and must correspond to the explanations and results presented in the report.

This section must include **Table 7: Implemented or Modified Project Files**. The table must list each important file, its type, and its purpose in the implementation.

This section must also include **Table 8: OMNeT++, INET, and FLoRa Usage**. The table must state how each framework is used in the submitted project.

Table 7: Implemented or Modified Project Files

File Name	File Type	Purpose
	NED	
	C++	
	Header	
	INI	
	MSG	

Table 8: OMNeT++, INET, and FLoRa Usage

Tool/Framework	How It Is Used in the Project
OMNeT++	
INET	
FLoRa	

6.8 Mapping of Course Topics to the Project

This section must include **Table 9: Mapping of CNG 476 Course Topics to the Final Project**. The table must explain how each course topic is implemented in the submitted project. Generic statements are not sufficient.

6.9 Experimental Design

This section must describe how the experiments were conducted. It must include the simulation duration, number of simulation runs, random seeds, baseline scenario, alternative scenarios, varied parameters, and collected metrics.

This section must include **Table 10: Experimental Design Summary**. The table must contain the main experimental settings used to generate the final results.

6.10 Performance Metrics

This section must define the performance metrics used in the project. Each metric used in the results must be clearly defined before it appears in the results section.

This section must include **Table 11: Performance Metrics**. The table must list each metric, its unit, and its definition or purpose.

6.11 Input Analysis

This section must justify the simulation inputs. It must include at least one of the following: histogram, QQ-plot, fitted distribution, trace-based explanation, or mathematical justification of assumed input parameters.

This section must include **Table 12: Input Analysis Summary**. The table must identify each input variable, the analysis method used, and the conclusion obtained from the analysis.

Table 9: Mapping of CNG 476 Course Topics to the Final Project

Course Topic	Project-Specific Implementation
Simulation methodology and modelling	Explain entities, assumptions, resources, events, and objectives.
Random number generation	Explain where pseudo-random numbers are used.
Monte Carlo simulation	Explain repeated runs, seeds, and scenario repetitions.
Probability models	State and justify the probability distributions used.
Inverse transformation or sampling	Explain at least one non-uniform sampling method, if used.
Stochastic processes	Explain arrivals, sensing, failures, reporting, service, or alarms.
Queueing analysis	Identify and analyse at least one queueing point.
Discrete-event simulation	Explain event-driven state changes.
Event scheduling	List scheduled events such as packet generation, service, timeout, or alarms.
Time-stepped / trace-driven view	Explain why DES is used and whether trace/time-stepped inputs are relevant.
Wireless network modelling	Explain nodes, links, coverage, interference, gateway access, or packet delivery assumptions.
OMNeT++ implementation	Explain the OMNeT++ modules and configuration.
INET integration	Explain the INET components or protocol support used.
FLoRa integration	Explain the LoRa/FLoRa communication model used.
Input analysis	Include histograms, QQ-plots, fitted distributions, traces, or parameter justification.
Output analysis	Include averages, variability, plots, and confidence intervals.
IoT application relevance	Explain the selected IoT domain and its importance.

6.12 Results and Output Analysis

This section must present and analyse the simulation results. It must include at least two scenarios or configurations, at least three performance metrics, tables or plots of results, average values, variability such as standard deviation, confidence interval for at least one main metric, and explanation of the observed trends.

The reported results must be generated from the submitted working implementation. Results that cannot be traced to the submitted project files may be treated as invalid.

This section must include **Table 13: Simulation Results Summary**. The table must report the scenario, metric, average value, and standard deviation or confidence interval.

Table 10: Experimental Design Summary

Item	Value / Description
Simulation Duration	
Number of Runs	
Random Seeds	
Baseline Scenario	
Alternative Scenarios	
Varied Parameters	
Collected Metrics	

Table 11: Performance Metrics

Metric	Unit	Definition / Purpose
Packet Delivery Ratio	%	
End-to-End Delay	s or ms	
Queue Length	packets	
Throughput	packets/s	
Energy Consumption	J	

6.13 Discussion

This section must discuss the meaning of the results. It must explain the observed trade-offs, limitations, and whether the results match the expected behaviour of the system.

This section must include **Table 14: Discussion of Main Findings**. The table must list the main findings and provide a technical explanation for each finding.

6.14 Conclusion

The conclusion must summarise what was implemented, what was evaluated, the main findings, the main limitation, and one possible future extension.

6.15 References

The final report must include a references section with at least **15 valid references**. The references must be relevant, technically appropriate, and cited inside the report text.

The references must follow IEEE citation style. Each reference must be cited in the main text of the report. A source that appears only in the reference list but is not cited inside the report will not be counted toward the minimum requirement.

The references section must not be filled with irrelevant, duplicated, incomplete, or non-technical sources. General websites, blogs, and unverified online material should not be used unless there is a clear technical reason.

Table 12: Input Analysis Summary

Input Variable	Analysis Method	Conclusion
Inter-arrival Time	Histogram / QQ-plot / Distribution	
Service Time	Histogram / Distribution	
Alarm/Fault Event	Probability model / Trace	

Table 13: Simulation Results Summary

Scenario	Metric	Average	Std. Dev. / CI
Baseline			
Scenario 1			
Scenario 2			

6.16 Code Availability

This section must provide the public GitHub repository link. The following format may be used:

> The complete working project source code, simulation files, configuration files, result-processing scripts, and final report PDF are publicly available at: `https://github.com/username/repository-name`

6.17 Individual Contributions

This section must include **Table 15: Individual Contributions**. The table must state the name, student ID, and contribution of each group member.

7 Final Report Grading Rubric: 50%

The final report will be graded out of **50 marks**.

Table 16: Final Report Grading Rubric

Criterion	Marks	Required Evidence
Format, first-page information, and completeness	4	IEEE format, group details, course details, GitHub link.
Abstract, introduction, and project description	5	Clear problem, motivation, objectives, selected scenario, and project summary table.

Criterion	**Marks**	**Required Evidence**
System architecture	5	Technical figure, system component table, and explanation of system components.
System model and assumptions	6	Entities, events, states, queues, traffic model, assumptions, and required model tables.
Simulation methodology	6	DES logic, RNG, probability models, stochastic processes, Monte Carlo runs, and event tables.
OMNeT++, INET, and FLoRa implementation	7	Actual files/modules/configurations, message flow, framework usage table, and evidence of correct execution.
Mapping to course topics	5	All major course topics mapped to the implemented project in table form.
Experimental design and performance metrics	4	Runs, seeds, scenarios, varied parameters, metrics table, and experimental design table.
Input analysis	3	Histogram, QQ-plot, fitted distribution, trace, justified parameters, and input analysis table.
Results and output analysis	7	Scenarios, metrics, plots/tables, averages, variability, confidence interval, and results summary table.
Discussion, conclusion, references, and contributions	3	Interpretation, limitations, future work, at least 15 valid cited references, and individual contribution table.
Total	**50**	

Table 14: Discussion of Main Findings

Finding	Explanation

Table 15: Individual Contributions

Name	Student ID	Contribution
Student 1	ID	
Student 2	ID	
Student 3	ID	

8 GitHub Repository Requirements and Rubric: 10%

The GitHub repository must be public and must contain the complete working project. The repository must include all files required to compile, run, and reproduce the main simulation results.

The repository must include:

1. Complete OMNeT++ project files.
2. NED files.
3. C++ source/header files, if developed.
4. Message files, if used.
5. INI configuration files.
6. Scripts used for result processing or plotting, if any.
7. Result files or sample output files.
8. Final report PDF.
9. README file explaining how to set up and run the project.

9 Final Demo Rubric: 20%

The final demo will be graded out of **20 marks**. The group must demonstrate a working version of the submitted project. The demo must show that the project can be executed correctly and that the reported results are based on the submitted implementation.

All group members must be able to explain their contributions. Individual marks may be adjusted if a student cannot explain the submitted work.

Table 17: GitHub Repository Rubric

Criterion	**Marks**
Repository is public and accessible	2
Complete working project source code is uploaded	3
README explains setup and execution steps	2
Report PDF is included in the repository	1
Results/scripts/configuration files are included	1
Repository is organised and understandable	1
Total	**10**

10 Minimum Conditions

A project may receive a significantly reduced grade if any of the following conditions are not satisfied:

1. The report PDF is not submitted to ODTU CLASS.
2. The GitHub repository is missing, private, or inaccessible.
3. The submitted project is not in working condition.
4. The project cannot compile or execute without reasonable explanation.
5. The project does not use OMNeT++.
6. INET and FLoRa are not meaningfully used.
7. The report does not include valid simulation results.
8. The reported results are not generated from the submitted implementation.
9. The report does not include at least 15 valid cited references.
10. The group does not attend the final demo.
11. The report exceeds the accepted Turnitin similarity/AI threshold without satisfactory explanation.

11 Recommended IEEEtran Report Skeleton

```
\documentclass[conference]{IEEEtran}

\usepackage{amsmath,amssymb}
```

Table 18: Final Demo Rubric

Criterion	Marks
Clear explanation of project problem and motivation	2
Explanation of system architecture and simulation model	3
Code walkthrough: NED, C++, INI, and relevant files	4
Simulation executes correctly during demo or is demonstrated with valid evidence	4
Explanation of OMNeT++, INET, and FLoRa usage	3
Explanation of results and performance metrics	3
Individual oral understanding of group members	1
Total	**20**

```
\usepackage{graphicx}
\usepackage{booktabs}
\usepackage{url}
\usepackage{cite}

\begin{document}

\title{Project Title: CNG 476 System Simulation Final Report}

\author{
\IEEEauthorblockN{Student Name 1, Student Name 2, Student Name 3}
\IEEEauthorblockA{
Computer Engineering Program\\
Middle East Technical University Northern Cyprus Campus\\
Student IDs: ID1, ID2, ID3\\
GitHub: https://github.com/username/repository-name}
}

\maketitle

\begin{abstract}
...
\end{abstract}
```

```
\begin{IEEEkeywords}
System simulation, OMNeT++, INET, FLoRa, IoT,
discrete-event simulation.
\end{IEEEkeywords}

\section{Introduction}
\section{Selected Project Description}
\section{System Architecture}
\section{System Model and Assumptions}
\section{Simulation Methodology}
\section{OMNeT++, INET, and FLoRa Implementation}
\section{Mapping of Course Topics to the Project}
\section{Experimental Design}
\section{Performance Metrics}
\section{Input Analysis}
\section{Results and Output Analysis}
\section{Discussion}
\section{Conclusion}
\section*{Code Availability}
\section*{Individual Contributions}

\bibliographystyle{IEEEtran}
\bibliography{references}

\end{document}
```